

Exp No: 24 MODELING CLOCK-DRIVEN CONCURRENT PROCESSES USING SC_THREAD AND WAIT() IN SYSTEMC

Program:

```
// Code your testbench here.
// Uncomment the next line for SystemC modules.
// #include <systemc>
#include <systemc.h>

SC_MODULE(ClockExample) {
    sc_in<bool> clk;
    void process() {
        while (true) {
            wait();
            std::cout << "Time: " << sc_time_stamp()
                << " - Clock Triggered" << std::endl;
        }
    }
}

SC_CTOR(ClockExample) {
    SC_THREAD(process);
    sensitive << clk.pos();
}

};

int sc_main(int argc, char* argv[]) {
    sc_clock clock("clock", 1, SC_NS);
    ClockExample module("module");
    module.clk(clock);
    sc_start(5, SC_NS);
    return 0;
}
```

New Run Save Copy

testbench.cpp

```
1 // Code your testbench here.
2 // Uncomment the next line for SystemC modules.
3 // #include <systemc>
4 #include <systemc.h>
5
6 SC_MODULE(ClockExample) {
7     sc_in<bool> clk;
8     void process() {
9         while (true) {
10             wait();
11             std::cout << "Time: " << sc_time_stamp()
12                 << " - Clock Triggered" << std::endl;
13         }
14     }
15
16     SC_CTOR(ClockExample) {
17         SC_THREAD(process);
18         sensitive << clk.pos();
19     }
20 };
21
22 int sc_main(int argc, char* argv[]) {
23     sc_clock clock("clock", 1, SC_NS);
24     ClockExample module("module");
```

design.cpp

Log Share

```
[2026-08-21 04:18:47 UTC] g++ -o sim -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...

SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
Copyright (c) 1996-2018 by all Contributors,
ALL RIGHTS RESERVED
Time: 0 s - Clock Triggered
Time: 1 ns - Clock Triggered
Time: 2 ns - Clock Triggered
Time: 3 ns - Clock Triggered
Time: 4 ns - Clock Triggered
Done
```